Pricing Off Accounting Costs

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Pricing and Inventory Costs: Fed Beige Book

- "Several contacts noted they were still working through pre-tariff inventories, thus delaying price adjustments."

  — Federal Reserve Bank of Atlanta, June 2025

- "Several contacts across different retail sectors phased in price increases as old inventories were sold and replaced with goods subject to tariffs. Retailers expect additional price increases as this phase-in continues."

  — Federal Reserve Bank of San Francisco, Sep 2025

Pricing and Inventory Costs: Fed Beige Book

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  — Federal Reserve Bank of San Francisco, Sep 2025

- "A coffee roaster noted that while tariffs on coffee have largely been lifted, selling prices will only go down once the stock of inventory acquired at higher costs has been cleared."

  — Federal Reserve Bank of New York, Jan 2026

Pricing and Inventory Costs: Earnings Calls

- Manufacturers: "We have the opportunity for those duties that we pay, those tariffs that we pay, to recover that when we deliver the aircraft. Now, you know, it creates a little bit of a cash flow timing issue that we'll have to work through."

  — Boeing, Apr 23 2025

- Distributors: "We anticipate implementing additional pricing actions as inventory affected by tariffs moves through our cost of sales."

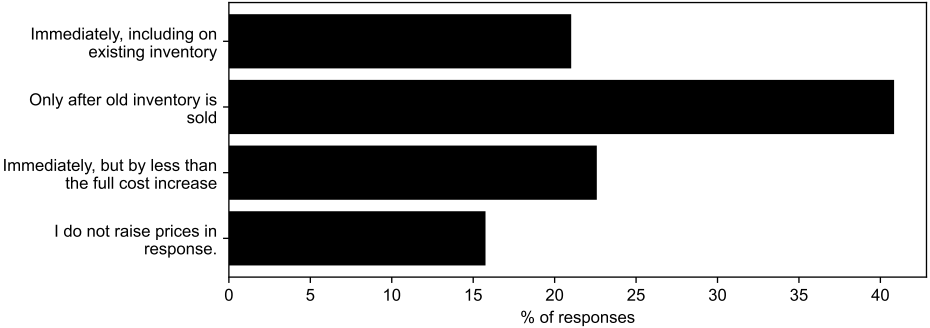
  — Global Industrial, July 29 2025

- Retailers: "Our pricing algorithms seek to maintain a balance between factoring in our higher wholesale costs and keeping us market priced. In practice, this means that for a short period of time, inventory-carrying retailers may have a slight cost advantage. This should dissipate over the next couple of quarters as the market rebalances, pre-tariff inventory is depleted, and all retailers incur higher wholesale costs reflective of tariffs."

  — Wayfair, Oct 31 2019

Pricing and Inventory Costs: Surveys

- "Suppose your input cost rises today, but you still have inventory bought at lower cost. When do you raise your sales price?"



Note: Survey of 400 firm price-setters by Schoenle (2026).

Pricing and Inventory Costs

- Standard theory: price off marginal cost $=$ current replacement cost.
- Historical purchase cost is sunk and thus irrelevant!
- Gordon (1981): "Many firms, however, appear to base prices on actual costs paid in the past, so-called 'FIFO' (first-in-first-out) pricing practices."

Two Propositions

1. Firms use backward-looking cost measures ("accounting costs").

   - "The need for objective and systematic measurement forces the cost standard to be backward-looking—or at least sideward- rather than forward-looking."

     — Okun (1981)

2. Firms largely set prices using these cost measures, rather than replacement costs.

   - "The reliance of firms on historical costs is evident in many empirical studies.... It is reasonably predictable that today's changes in cattle prices will show up on the supermarket counter only after a lag; the butcher does not keep changing price tags in pace with changes in some putative replacement cost."

     — Okun (1981)

Roadmap

1. Case study: Airlines.
2. Principal–agent model of pricing off accounting costs.
3. Price dynamics resemble sticky-price models.
   - But delays due to production lags / inventory stocks, not length of price spells.
   - Anticipation can delay pass-through (stocking up), opposite sticky-price models.
   - Measured markups can be acyclical/procyclical even if true markup countercyclical.
4. Quantitative results.
   - Cost shocks to goods-producing industries take avg. 5–6 months to show up in CPI.
   - Response of consumer prices to identified oil shocks, 2025 tariffs.

Selected Related Literature

- **Cost-based pricing**: Gordon (1981), Okun (1981), Blinder et al. (1998).
  - *Firms inattentive to future costs*: Gordon (1990), Mankiw and Reis (2002), Woodford (2003), Minton and Wheaton (2022), Afrouzi et al. (2024).

    $\Rightarrow$ Not inattention: (1) Forward buying, (2) Prices don't change even when buying at new price.
- **Other pricing puzzles**:
  - *Full-cost pricing*: E.g., Hall and Hitch (1939), Altomonte et al. (2015), Bewley (2025).
  - *Response to cost vs. demand shocks*: Cagan (1979), Bils and Chang (2000), Fabiani et al. (2006), Gagnon and Lopez-Salido (2020), Kohler et al. (2026).
- **Verifiable info and firm monitoring**: E.g., Meckling and Jensen (1976), Holmström (1979), Graham et al. (2005), Rogerson (2008, 2011).
- **Fairness**: Kahneman et al. (1986), Rotemberg (2005, 2011), Eyster et al. (2021).

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Review of accrual accounting (FASB and IFRS)

- Matching principle: Match timing of expenses to revenues.
- Cost principle: Value expenses at historical cost.

Expense Recognition (Matching) Principle

The expense recognition principle (also referred to as the matching principle) states that we must match expenses with associated revenues in the period in which the revenues were earned. A mismatch in expenses and revenues could be an understated net income in one period with an overstated net income in another period. There would be no reliability in statements if expenses were recorded separately from the revenues generated.

For example, if Lynn earned printing revenue in April, then any associated expenses to the revenue generation (such as paying an employee) should be recorded on the same income statement. The employee worked for Lynn in April, helping her earn revenue in April, so Lynn must match the expense with the revenue by showing both on the April income statement.

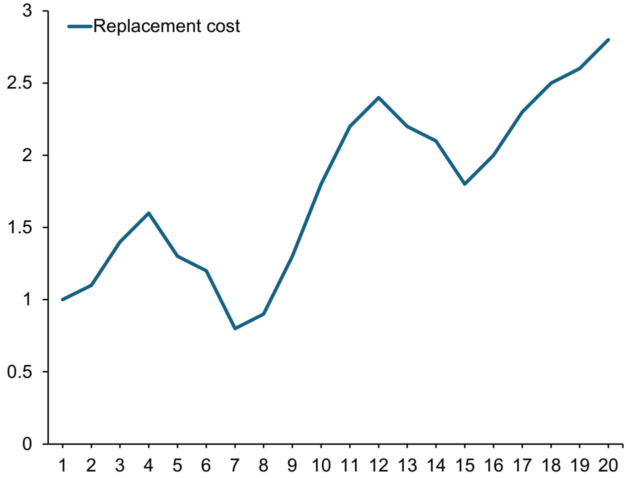
Cost Principle

The cost principle, also known as the historical cost principle, states that virtually everything the company owns or controls (*assets*) must be recorded at its value at the date of acquisition. For most assets, this value is easy to determine as it is the price agreed to when buying the asset from the vendor. There are some exceptions to this rule, but always apply the cost principle unless FASB has specifically stated that a different valuation method should be used in a given circumstance.

The primary exceptions to this historical cost treatment, at this time, are financial instruments, such as stocks and bonds, which might be recorded at their fair market value. This is called mark-to-market accounting or fair value accounting and is more advanced than the general basic concepts underlying the introduction to basic accounting concepts; therefore, it is addressed in more advanced accounting courses.

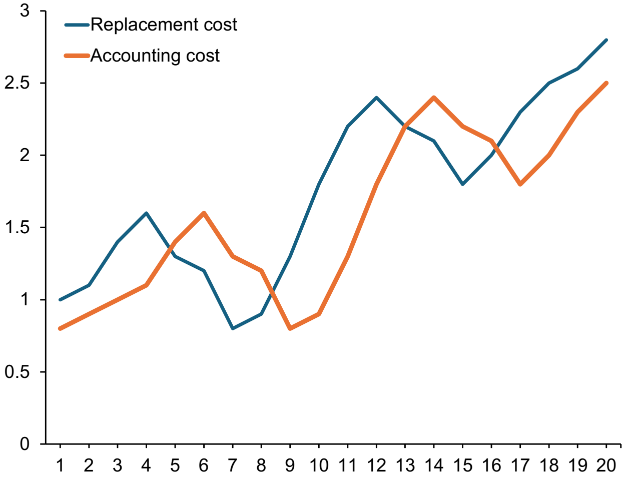
Once an asset is recorded on the books, the value of that asset must remain at its historical cost, even if its value in the market changes. For example, Lynn Sanders purchases a piece of equipment for $40,000. She believes this is a bargain and perceives the value to be more at $60,000 in the current market. Even though Lynn feels the equipment is worth $60,000, she may only record the cost she paid for the equipment of $40,000.

Example: Replacement vs. accounting costs



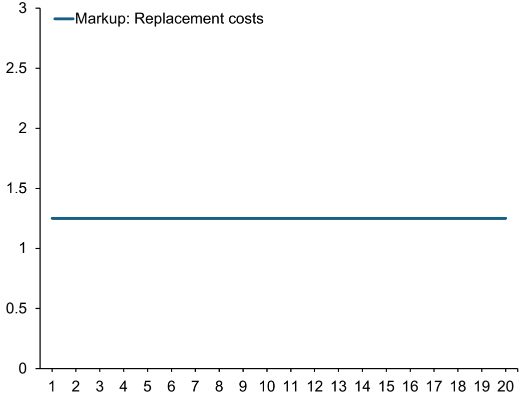
- Replacement costs c_t.
- Inventory stock I_t, units sold D_t.
- Example: D and I constant,
  $\delta = D/I = 0.5$.
  (Two months of inventory.)

Example: Replacement vs. accounting costs



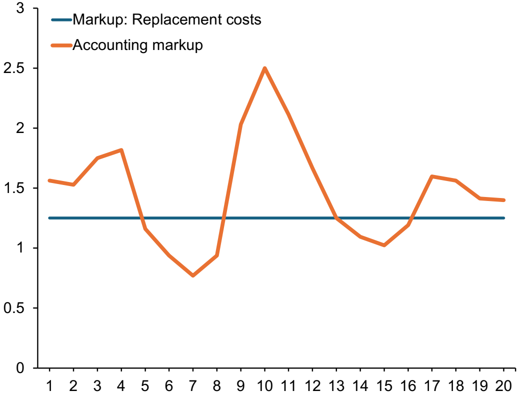
- Replacement costs c_t.
- Inventory stock I_t, units sold D_t.
- Example: D and I constant, $\delta = D/I = 0.5$. (Two months of inventory.)
- First-in-first-out (FIFO) accounting.
- Cost of goods sold (COGS) $= h_t D_t$, where $h_t = c_{t-2}$.

Example: Accounting markups



- Suppose optimal price $p_t = 1.25 \times c_t$.

Example: Accounting markups



- Suppose optimal price $p_t = 1.25 \times c_t$.
- Sales / COGS $= 1.25 \times c_t / h_t$.
  $\Rightarrow$ Volatile accounting margins.

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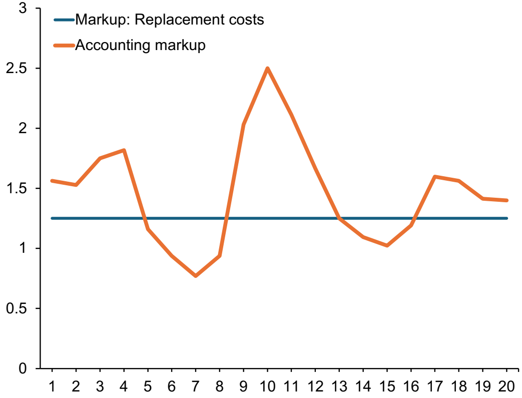

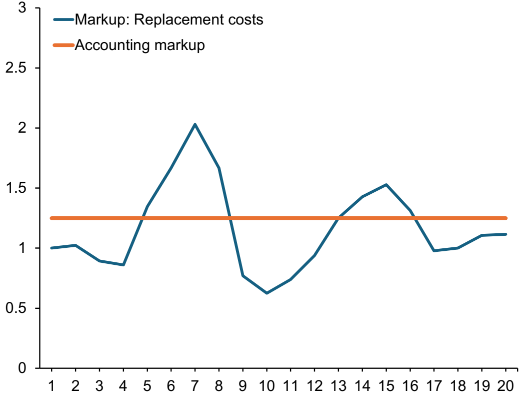

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- Sales / COGS $= 1.25 \times c_t / h_t$.
  $\Rightarrow$ Volatile accounting margins.

- If $p_t = 1.25 \times h_t$ (price off acct cost), replacement cost markups more volatile.

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Airlines: Accounting costs for jet fuel

CFO JOURNAL

Why Southwest Airlines Finally Soured on Jet Fuel Hedging

The company, an outlier in the airline industry for its commitment to hedging, says the program got too costly

By Kristin Broughton
March 21, 2025 3:49 pm ET

BUSINESS • AIRLINES

Delta's Ace in the Hole for Surging Jet Fuel Costs: Its Own Refinery

Airline's 2012 acquisition of Pennsylvania refinery has been called both prescient and pointless

By Alison Sider
April 9, 2026 5:30 am ET

- For hedgers, accounting cost reflects fuel price incl. hedging gains/losses.

Airlines: Accounting costs for jet fuel

- Jet fuel costs constitute 20–30% of airline operating costs.
- Airlines face same fuel spot price, but diff. accounting costs from hedging gains/losses.
- Data from Bureau of Transportation Statistics (BTS) on 10 airlines: American, Alaska, Allegiant, Delta, Frontier, Hawaiian, JetBlue, Southwest, Spirit & United.
    - Accounting costs of fuel: BTS Form 41 Schedule P-12(a).
    - Spot price of jet fuel: Energy Information Administration.
    - Ticket prices: DB1B (10% sample of all itineraries).

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1. Accounting costs of fuel are backward-looking for hedgers.
2. Sluggish adjustment of accounting cost $\Rightarrow$ sluggish adjustment of fares.

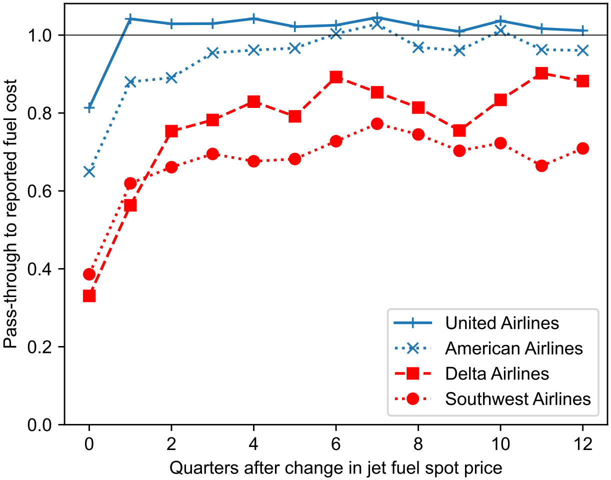
Airlines: Accounting costs for jet fuel

- Delta (Dec 2008 10K): “Throughout the summer months, fuel prices remained at record high levels... However, fuel prices fell dramatically during the third and fourth quarters, creating sizable losses on our fuel hedge contracts in the fourth quarter. Losses on our derivative contracts that relate to jet fuel purchases in 2009 are deferred on our Consolidated Balance Sheet for 2008 and will be recognized in 2009 when the hedged jet fuel is purchased and consumed.”

	1Q	**2Q**	**3Q**	**4Q**	**2009**
Call options	7%	12%	29%	19%	17%
Collars	33%	26%	4%	0%	16%
Swaps	37%	37%	23%	17%	28%
Total	77%	75%	56%	36%	61%
Downside participation		37%	73%	83%	
All-in projected fuel price / gallon*	$2.26	$2.08	$1.91	$1.73	$1.99
Hedge loss/gallon included in projected price	$0.71	$0.45	$0.18	$0.03	$0.34

* Includes tax and transportation costs of approximately $0.17/gallon, and call option premiums

Airlines: Sluggish adjustment of accounting cost to spot price



- Pass-through of jet fuel spot price to fuel accounting costs.

$$\Delta c_{it}^{\text{accounting}} = \alpha_i + \sum_{k=0}^{12} \beta_{ik} \Delta c_{t-k}^{\text{spot}} + \varepsilon_{it}.$$

- For hedgers, delayed pass-through to accounting measure of cost.
  - Southwest: 0.70 at one year.
  - Delta: 0.78 at one year.

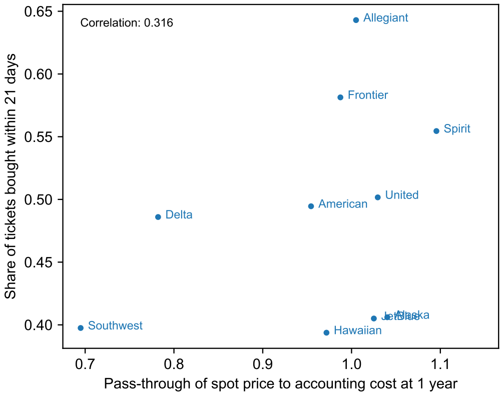
Airlines: Prices set off accounting costs

- Airlines' accounting costs of fuel predict fares (Fuel Cost Diff$_{it}$ = Log Carrier i's Reported Fuel Cost$_{it}$ − Log Spot Price$_t$).
- Similar if isolate variation in accounting cost from hedging $\Rightarrow$ backward-looking costs.

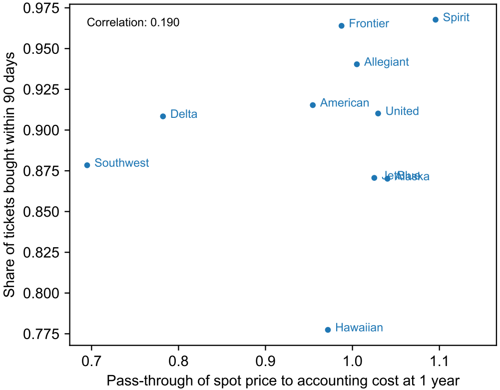
	Log Fare$_{irt}$				Fuel Cost Diff$_{it}$
	OLS	OLS	RF	IV	FS
	(1)	(2)	(3)	(4)	(5)
Log Spot Fuel Price$_t$	0.317**	0.348**	0.332**	0.354**	
	(0.030)	(0.028)	(0.030)	(0.030)	
Fuel Cost Diff$_{it}$		0.522**		0.613**	
		(0.106)		(0.184)	
$\Delta_{t,t-4}$ Log Spot Price $\times$ Hedge Ratio$_i$			−0.593**		−1.018**
			(0.192)		(0.127)
Carrier $\times$ Route FEs	Yes	Yes	Yes	Yes	Yes
N	1765536	1765168	1765536	1765168	1765168
R^2	0.57	0.58	0.58	0.58	0.37

Airlines: Sticky prices?

- Concern: Hedging behavior correlated with share of customers that book flights early?
- DB1C data from May 2025–Aug 2026 shows correlation is positive, but noisy.



(a) Share bought $\leq$ 3 weeks in advance.



(b) Share bought $\leq$ 3 months in advance.

Airlines: Sticky prices, asymmetry, strategic complementarities...

	Log Fare$_{irt}$				
	Strategic complem.		Asymmetry	Sticky fares	
	(1)	(2)	(3)	(4)	(5)
Log Spot Fuel Price$_t$	0.348**	0.350**	0.354**	0.356**	0.355**
	(0.028)	(0.028)	(0.030)	(0.030)	(0.030)
Fuel Cost Diff$_{it}$	0.373**	0.436**		0.408**	0.409**
	(0.136)	(0.107)		(0.115)	(0.115)
Passenger Share on Route$_{it}$	−0.100**				
	(0.017)				
Fuel Cost Diff$_{it}$ × Passenger Share$_{it}$	0.230**				
	(0.108)				
Fuel Cost Diff for All Airlines on Route$_{rt}$		0.104**			
		(0.038)			
Fuel Cost Diff$_{it}$ (Positive)			0.653**		
			(0.168)		
Fuel Cost Diff$_{it}$ (Negative)			0.360**		
			(0.174)		
$\Delta_{t,t-4}$ Log Spot Price × Share Tickets Bought $\geq$3 Weeks in Advance$_i$				−0.153*	
				(0.081)	
$\Delta_{t,t-4}$ Log Spot Price × Share Tickets Bought $\geq$3 Months in Advance$_i$					−0.801*
					(0.434)
Carrier × Route FEs	Yes	Yes	Yes	Yes	Yes
N	1765168	1765168	1765168	1765168	1765168
R^2	0.58	0.58	0.58	0.58	0.58

Airlines: Volatility of accounting margins

- If firms price off replacement cost, gap between replacement vs. accounting cost creates volatility in accounting margins. $\Rightarrow$ Correcting for gap should reduce volatility.
- Instead, correcting for gap raises volatility, esp. for hedgers.

Carrier	Pass-through at 1 year	Volatility of operating margins: Reported	Volatility of operating margins: Corrected	Pct. change
Southwest	0.70	0.064	0.084	+31.2
Delta	0.78	0.079	0.087	+10.4
American	0.95	0.067	0.071	+5.4
Hawaiian	0.97	0.108	0.110	+2.0
Frontier	0.99	0.171	0.170	−0.5
Allegiant	1.01	0.081	0.079	−2.4
JetBlue	1.03	0.078	0.085	+10.2
United	1.03	0.072	0.072	−0.07
Alaska	1.04	0.098	0.101	+3.3
Spirit	1.10	0.136	0.135	−0.8

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Why price off accounting costs?

- “The need for objective and systematic measurement forces the cost standard to be backward-looking—or at least sideward- rather than forward-looking. [...] Any attempt to forecast input costs or to put them on a current replacement basis, which necessarily entails a forecast, would add to expenses, introduce a subjective element into cost calculations, and complicate the task of managerial control. Bygones cannot be mere bygones if they are the sole sources of systematic information.”

— Okun (1981), *Prices and Quantities*

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1. Ingredient #1: Replacement cost unverifiable, but accounting metrics observable.
2. Ingredient #2: Misaligned incentives between principal and manager.

Two-period model: Frictionless

$t = 0$
Manager purchases
I units of inventory
at price h.

$t = 1$
Manager observes c
and sets output price p.
Sales $pD(p)$.

$t = 2$
Manager liquidates
leftover inventory
at price c.

- No discounting. Assume I fixed and $D(p) < I$ for all possible p (no stockouts).
- $V^{\text{FB}} = \max_p \{-hI + pD(p) + c(I - D(p))\}$.
- $p^* = \arg\max_p (p - c)D(p)$.

Two-period model: Frictions

$t = 0$
Manager purchases
I units of inventory
at price h.

$t = 1$
Manager observes c and sets p.
Principal observes cost signal s.
Principal pays manager K.

$t = 2$
Manager liquidates
leftover inventory
at price c.

- Principal delegates firm operations to manager.
- Friction 1: Replacement cost c is manager's private information.
- Public signal $s \sim N(c/h, \tau^{-1})$. Friction disappears as precision $\tau \to \infty$.

Two-period model: Frictions

- Friction 2: Misaligned incentives.

$$V^{\text{manager}} = \max_{p} \Big\{ \underbrace{-hI + pD(p) + c(I - D(p))}_{\text{True firm value}} + \underbrace{(\alpha - 1)\,p}_{\text{Pricing wedge}} + \underbrace{K\,(pD(p), hD(p); s)}_{\text{Incentive payment}} \Big\}.$$

- When $\alpha \neq 1$, manager wants to set price lower or higher than optimal.

  E.g., Baumol (1959), Bertrand and Mullainathan (2003), Graham et al. (2005).

- Manager preference α is private information.

- Principal can incentivize manager, but can only condition payment on observed info.

Two-period model: Frictions

- Why can't principal condition payment on liquidation proceeds at $t = 2$?
  - Liquidation proceeds reveal c to both parties.
  - We assume that contract must be tied to performance observable at $t = 1$ $\Rightarrow$ need to monitor and incentivize manager in a timely manner.
- Why can't manager just tell principal c?
  - Manager always has incentive to misreport c in order to justify desired price.
- Principal's trade-off:
  - Stronger incentives limit extent to which manager preference α affects price...
  - ...but also lead manager to disregard valuable info about c.

Two-period model

- Log-linearize around $h = 1$ and $\alpha = 1$. Denote log-deviation of x by $\hat{x}$.
- Define $\sigma_\alpha^2 = \mathbb{E}[\hat{\alpha}^2]$ and $\sigma_c^2 = \mathbb{E}[(\hat{c} - \hat{h})^2]$.
- Local demand elasticity η. I.e., $\hat{D} = -\eta \hat{p}$.
- Assume principal's payment takes the form,

$$K(pD(p), hD(p); s) = -\frac{\kappa}{2}\left(\frac{pD(p)}{hD(p)} - \mu(s)\right)^2, \tag{1}$$

where $\mu(s)$ is a target accounting markup.

Pricing without incentives

Proposition (Pricing without incentives)

Suppose there is no incentive payment ($\kappa = 0$). Then, the price set by manager is

$$\hat{p} = \hat{c} + \psi_0\,\hat{\alpha}, \qquad \textit{where} \qquad \psi_0 = \frac{1}{\eta - 1}\left(\frac{\eta}{\eta - 1}\right)^{\eta}.$$

- $\psi_0 =$ elasticity of price to manager preference α when $\kappa = 0$.
  - Decreasing in demand elasticity η.
  - Managerial slack harder to sustain in competitive industries.
    Hart (1983), Giroud and Mueller (2010, 2011).

Optimal contract parameters

Proposition (Principal's optimal contract)

Given quadratic incentive payment (1)*, the principal's optimal choice of* $\mu(s)$ *and* κ *are*

$$\underbrace{\mu^*(s) = \frac{\eta}{\eta-1}\exp\left(\frac{\tau\sigma_c^2}{1+\tau\sigma_c^2}s\right)}_{\text{Target accounting markup}}, \qquad \underbrace{\kappa^* = \frac{1}{\eta}\left(\frac{\eta}{\eta-1}\right)^{\eta}\frac{\sigma_\alpha^2}{\sigma_c^2}\left(1+\tau\sigma_c^2\right)}_{\text{Strength of incentives}}.$$

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- Principal "marks-to-market" inventory using signal s.
  - Publicly available signals (e.g., trend inflation) priced in.

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- Principal "marks-to-market" inventory using signal s.
  - Publicly available signals (e.g., trend inflation) priced in.
- Stronger incentives κ^* when...
  - Risk of discretion $\gg$ value of private info, $\sigma_\alpha^2/\sigma_c^2$.
  - Demand elasticity η low.
  - Cost observed precisely ($\tau\uparrow$).

Pricing under optimal contract

Proposition (Pricing with incentives)

Under payment's optimal contract, manager sets price

$$\hat{p} = \underbrace{\phi\,[\hat{c} + \psi\,\hat{\alpha}]}_{\substack{\textit{Replacement} \\ \textit{cost pricing}}} + \underbrace{(1-\phi)\,\hat{h}}_{\substack{\textit{Accounting} \\ \textit{cost pricing}}},$$

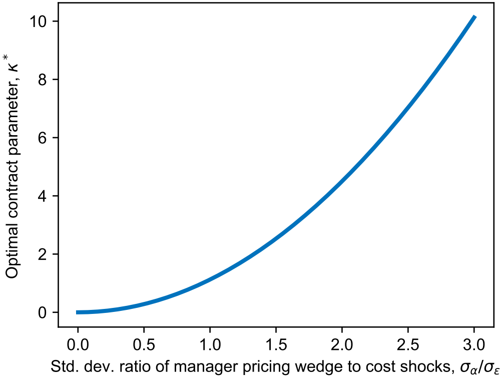
where

$$1-\phi = \frac{\sigma_\alpha^2/\sigma_c^2}{\sigma_\alpha^2/\sigma_c^2 + \psi_0^{-2} + \tau\sigma_\alpha^2}, \qquad \textit{and} \qquad \psi = \frac{1}{1+\tau\sigma_\alpha^2\psi_0^2}\psi_0.$$

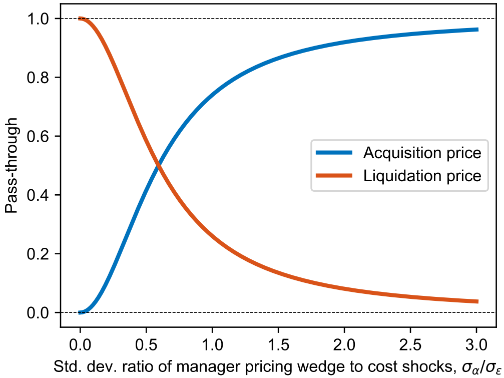
- Firm sets blended average of replacement cost and accounting cost price.
- $1-\phi$ is the extent of accounting cost pricing.
  - Increases with risk of managerial discretion ($\uparrow \sigma_\alpha^2/\sigma_c^2$).
  - Falls with competition ($\uparrow \eta$) and cost observability ($\uparrow \tau$).

Pricing under optimal contract

- As risk of manager discretion grows, κ^* rises and ϕ falls.

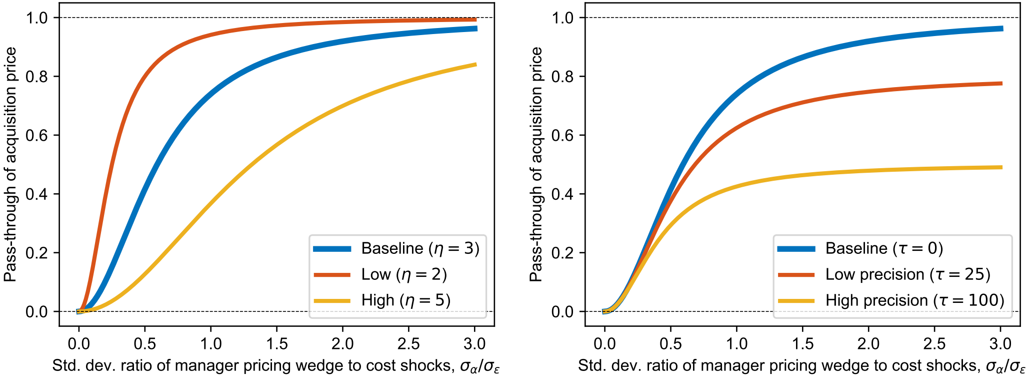


(a) Optimal contract parameter κ^*.



(b) Replacement vs. accounting cost pricing.

Comparative statics

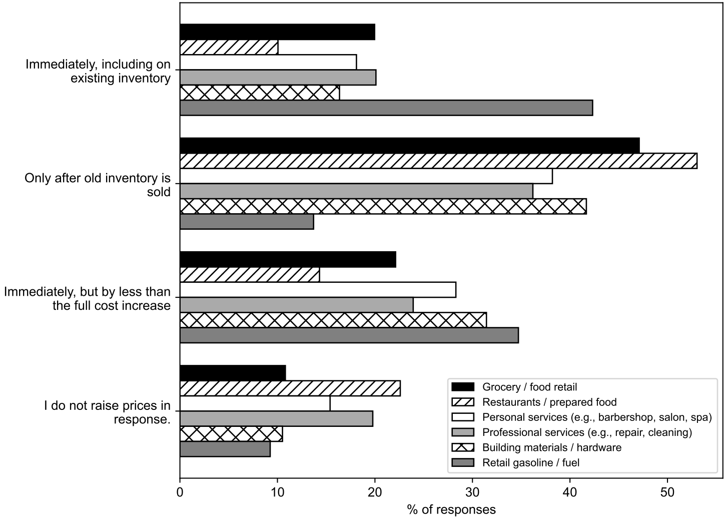


(a) Demand elasticity η.

(b) Signal precision τ.

- Better signal of costs or greater competition $\Rightarrow$ more weight on replacement cost.

Comparative statics



- Exception: gas stations.
- “The exceptions that occur to me arise when processed raw materials are sold on auction markets, as in the case of flour or soybean meal.” (Okun 1981).
- (Anecdotal) exception: sellers of gold/silver jewelry.

Other pricing puzzles

- Main result: Unverifiable info and principal–agent frictions mean:
  - Prices incorporate info irrelevant to p^{FB} (e.g., historical input prices).
  - Prices neglect info germane to p^{FB} but unverifiable (e.g., replacement costs).

Other pricing puzzles

- Main result: Unverifiable info and principal–agent frictions mean:
  - Prices incorporate info irrelevant to p^{FB} (e.g., historical input prices).
  - Prices neglect info germane to p^{FB} but unverifiable (e.g., replacement costs).
- Fixed costs: Suppose both c and overhead cost F are stochastic.
  - Pass through changes in fixed cost F like accounting cost $\hat{h}$ (weight $1-\phi$).
    Hall and Hitch (1939), Altomonte et al. (2015), Bewley (2025).
- Demand shocks: Suppose demand elasticity η is stochastic.
  - Underreact to changes in optimal markup like replacement cost $\hat{c}$ (weight ϕ).
    Cagan (1979), Bils & Chang (2000), Fabiani et al. (2006), Gagnon & Lopez-Salido (2020), Kohler et al. (2026).

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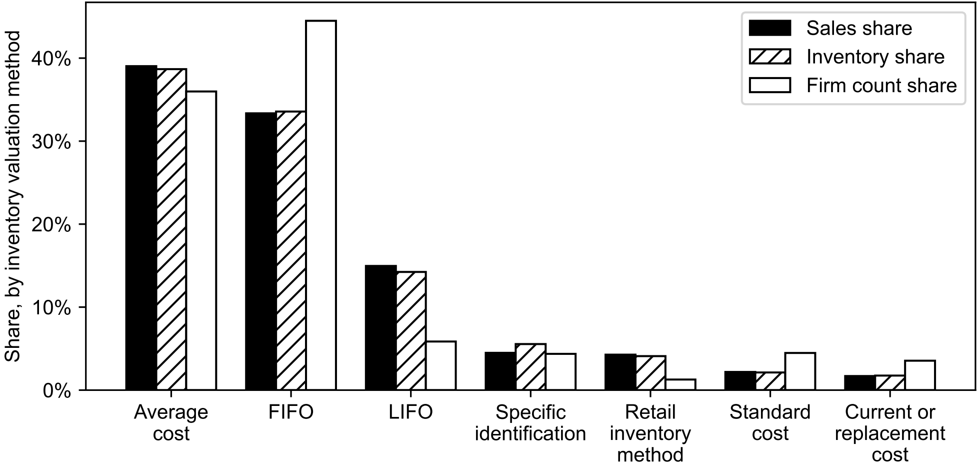
Calculating accounting costs

- In two-period setting, book value of inventory is unambiguous (hI).
- In dynamic setting, depends on how firm apportions cost across past purchases. E.g.,
  - Average acquisition cost (AAC).
  - First-in-first-out (FIFO).
  - Last-in-first-out (LIFO).
- Under all three, COGS reflects historical purchase costs. Methods differ in the weight they place on recent vs. more distant periods.
- $\Rightarrow$ Connection between accounting cost pricing and canonical sticky-price models.

AAC and FIFO most common inventory valuation methods

- Valuation methods used by inventory-carrying public firms over 2010–2025:



Price dynamics under average acquisition cost (AAC)

- Average acquisition cost (AAC) accounting:

$$h_t I_t = (I_{t-1} - D_t)\, h_{t-1} + Y_t c_t.$$

$\Rightarrow$ h_t is average of h_{t-1} and c_t, weighted by leftover inventory vs. new purchases.

- Log-linearizing around steady state with shipment-inventory ratio $\delta = D^{ss}/I^{ss}$,

$$\hat{p}_t = \hat{h}_t = (1-\delta)\,\hat{h}_{t-1} + \delta\,\hat{c}_t.$$

- Recursive formulation for prices looks familiar...!

Price dynamics under average acquisition cost (AAC) vs. Calvo

- Given a path of cost deviations from steady-state $\{\hat{c}_t\}_{t=-\infty}^{\infty}$,

$$\hat{p}_t^{\text{AAC}} = \delta \sum_{k=0}^{\infty} (1-\delta)^k \hat{c}_{t-k},$$

$$\hat{p}_t^{\text{Calvo}} = \theta \sum_{k=0}^{\infty} (1-\theta)^k \hat{c}_{t-k} + \underbrace{\theta\left[1-\beta(1-\theta)\right] \sum_{k=0}^{\infty} \sum_{s=0}^{\infty} \beta^s (1-\theta)^{k+s} \mathbb{E}_{t-k}\left[\hat{c}_{t+s-k} - \hat{c}_{t-k}\right]}_{\text{Due to forward-looking reset prices}},$$

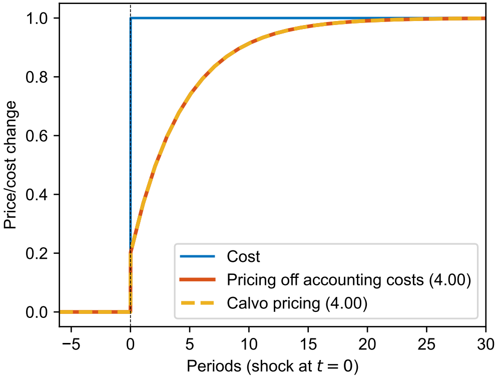
  - AAC: δ is steady-state shipment-inventory ratio.
  - Calvo: θ is fraction of firms able to reset prices each period, and β is discount factor.

- Paths coincide if $\delta = \theta$ and either $\beta = 0$ or $\mathbb{E}_{t+s}[\hat{c}_{t+s}] = \hat{c}_t$ for all t, s.

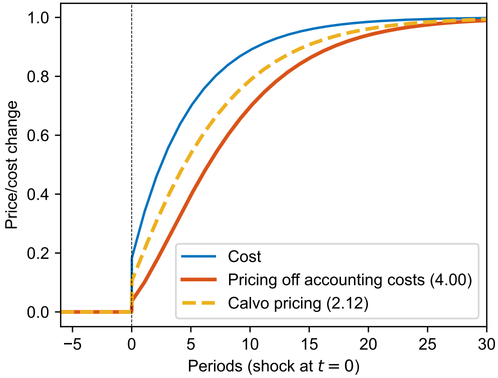
- Behaves like "myopic Calvo" (Minton and Wheaton 2022), but not due to inattention.

Example: More delays for slow-moving shocks

- Set $\delta = \theta = 0.2$. Shock path, unexpected before $t = 0$: $\hat{c}_t = 1 - \rho^{t+1}$.

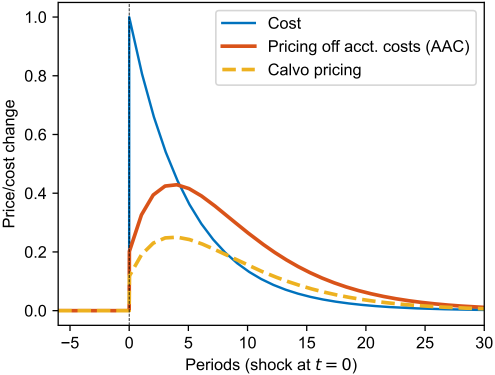


(a) Immediate shock ($\rho = 0$).

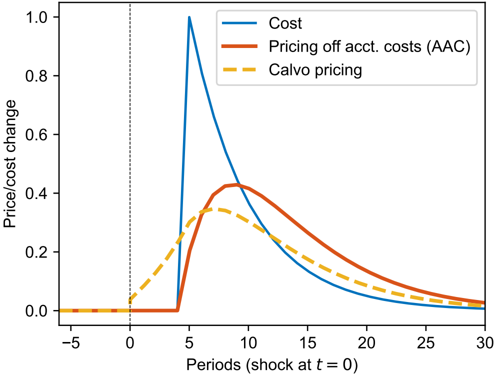


(b) Slow-moving shock ($\rho = e^{-0.2}$).

Example: More responsiveness to transitory shocks



(a) Unanticipated cost increase at $t = 0$.



(b) News at $t = 0$ about cost increase at $t = 5$.

- More volatility and less persistence of inflation. Bils and Klenow (2004), Bils et al. (2012).

FIFO and LIFO

- Given a path of cost deviations from steady-state $\{\hat{c}_t\}_{t=-\infty}^{\infty}$,

$$\hat{p}_t^{\text{FIFO}} = \hat{c}_{t-1/\delta},$$
$$\hat{p}_t^{\text{LIFO}} = \hat{c}_t,$$

FIFO and LIFO

- Given a path of cost deviations from steady-state $\{\hat{c}_t\}_{t=-\infty}^{\infty}$,

$$\hat{p}_t^{\text{FIFO}} = \hat{c}_{t-1/\delta},$$

$$\hat{p}_t^{\text{LIFO}} = \hat{c}_t,$$

$$\hat{p}_t^{\text{LIFO-}\delta} = \delta \sum_{k=0}^{1/\delta-1} \hat{c}_{t-k},$$

[Firms buy inventory every $1/\delta$ periods.]

FIFO and LIFO

- Given a path of cost deviations from steady-state $\{\hat{c}_t\}_{t=-\infty}^{\infty}$,

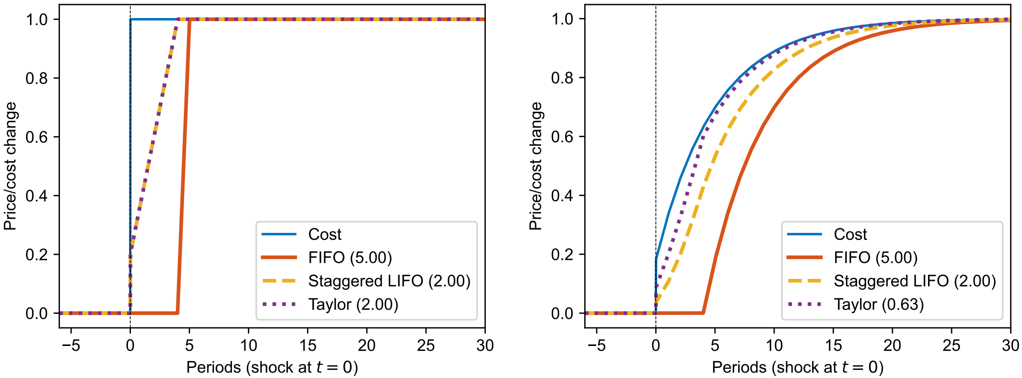
$$\hat{p}_t^{\text{FIFO}} = \hat{c}_{t-1/\delta},$$
$$\hat{p}_t^{\text{LIFO}} = \hat{c}_t,$$
$$\hat{p}_t^{\text{LIFO-}\delta} = \delta \sum_{k=0}^{1/\delta-1} \hat{c}_{t-k}, \qquad \text{[Firms buy inventory every } 1/\delta \text{ periods.]}$$
$$\hat{p}_t^{\text{Taylor}} = \frac{1}{N}\sum_{k=0}^{N-1} \hat{c}_{t-k} + \underbrace{\frac{1}{N}\frac{1-\beta}{1-\beta^N}\sum_{k=0}^{N-1}\sum_{s=0}^{N-1} \beta^s \mathbb{E}_{t-k}\left[\hat{c}_{t+s-k} - \hat{c}_{t-k}\right]}_{\text{Due to forward-looking reset prices}}.$$

  - LIFO w/ intermittent purchases (LIFO-δ): Firms stock up on inventory every $1/\delta$ periods.
  - Taylor: Firm resets price every N periods, and β is discount factor.

- LIFO-δ and Taylor coincide if $\delta = 1/N$ and either $\beta = 0$ or $\mathbb{E}_{t+s}[\hat{c}_{t+s}] = \hat{c}_t$ for all t, s.

Example: More delayed adjustment to slow-moving shocks

- Similar intuitions as AAC vs. Calvo.



(a) Immediate shock.

(b) Slow-moving shock.

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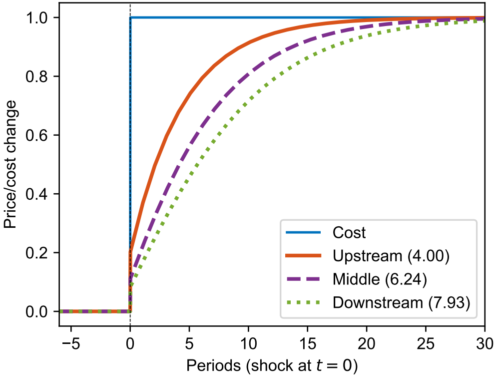
Implications vis-a-vis sticky-price models

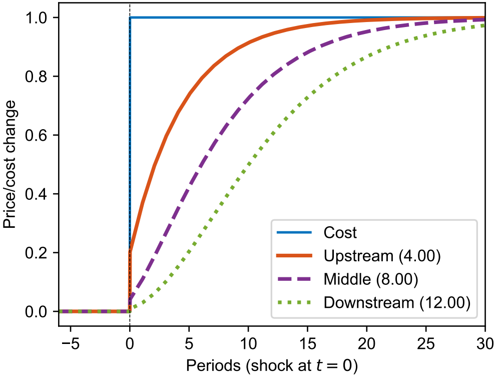
1. Supply chains amplify delays in price adjustment.
2. Anticipation of $\uparrow$ cost delays, rather than speeds up, pass-through if firms stock up.
3. Measured markups can be pro- or acyclical even when true markups countercyclical.

#1: Amplification of delays in supply chains

- Each firm turns cost shock into slower-moving shock for downstream firm.



(a) Calvo.



(b) Pricing off accounting costs (AAC).

#2: Forward buying delays pass-through of cost increases

- First-order approximation for prices did not require keeping track of inventory stocks:

$$\hat{p} = (1-\delta)\hat{h}_{t-1} + \delta\hat{c}_t.$$

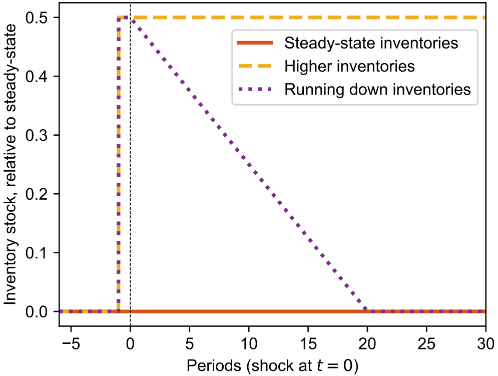
- Effects of stocking up on inventories show up at second-order:

$$\hat{h}_t = \underbrace{(1-\delta)\,\hat{h}_{t-1} + \delta\hat{c}_t}_{\text{First order}} + \underbrace{\frac{1}{2}\delta\,(1-\delta)\left(\hat{c}_t - \hat{h}_{t-1}\right)^2}_{\text{Second-order correction}} + \underbrace{\delta\left(\hat{c}_t - \hat{h}_{t-1}\right)\left(\hat{Y}_t - \hat{I}_t\right)}_{\text{Effect of inventories / purchases}}.$$

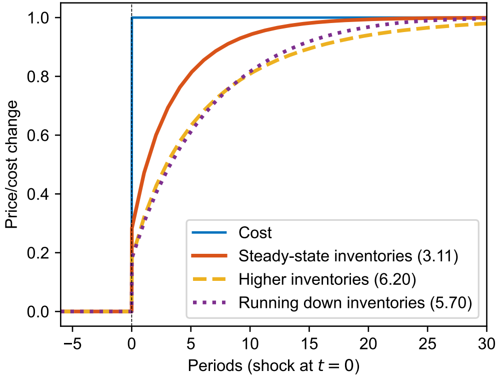
- If purchases low relative to inventory stock when replacement costs rise $\Rightarrow$ lower price.

#2: Forward buying delays pass-through of cost increases

- Illustrate with exogenous inventory paths in second-order approximation:



(a) Inventory paths.



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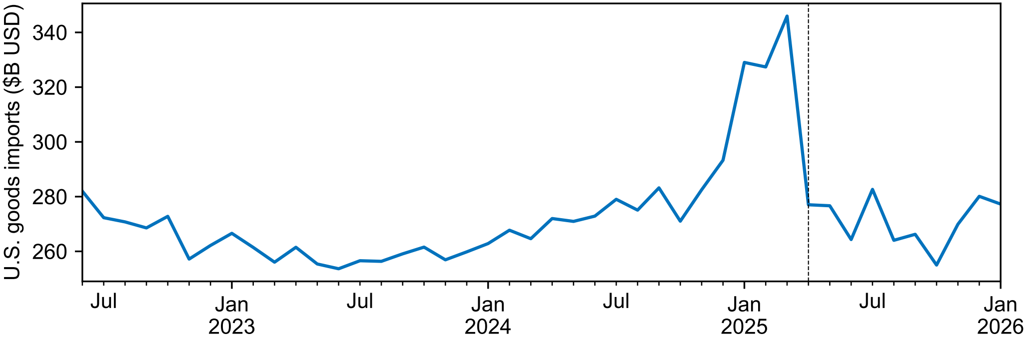
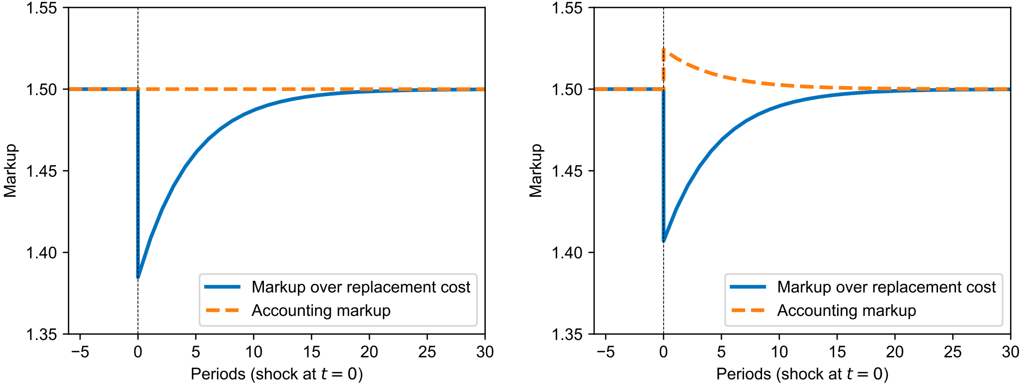


- Sticky-price models: Anticipation speeds up pass-through.
- Need “myopic expectations” to amplify delays. Minton and Wheaton (2022).
- But firms clearly stocking up in anticipation! Not myopic.

#3: Response of measured vs. true markups

- Debate: Markup cyclicality match NK model? Nekarda and Ramey (2020), Anderson et al. (2025).
- Measured markups can be flat / rise even if true markup vs. replacement cost falls.



(a) Full accounting cost pricing ($\phi = 0$).

(b) Partial accounting cost pricing ($\phi = 0.2$).

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Accounting cost pricing ⇒ macro rigidity?

- "It is reasonably predictable that today's changes in cattle prices will show up on the supermarket counter only after a lag; the butcher does not keep changing price tags in pace with changes in some putative replacement cost."

  — Okun (1981), *Prices and Quantities*

- "Industry experts estimate it takes at least nine months for raw bean prices to filter through to coffee drinkers [....] because U.S. roasters typically hold about two to three months' worth of bean stocks on average and need another two to three months to roast and package their products." — *Reuters, Dec 19 2025*

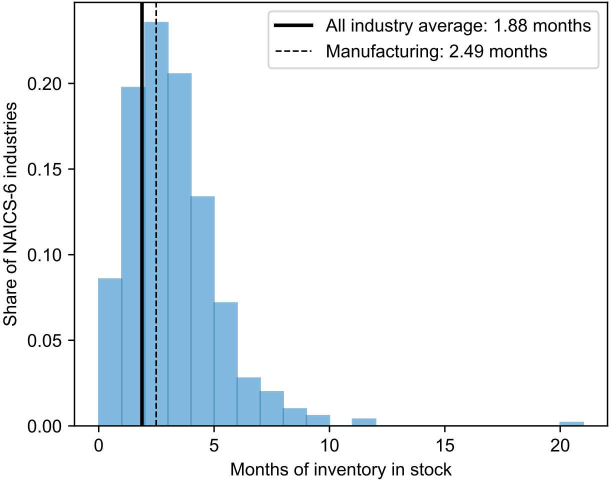
- "Cocoa—the key ingredient in chocolate—has been steadily falling in price from its all-time high in 2024. [...] But that drop hasn't trickled down to what consumers are paying. [...] The reason: The Easter chocolate on shelves today was produced using cocoa purchased at its extreme high." — *CNN Business, Apr 5 2026*

Quantitative input–output model of U.S. economy

Variable	Description	Source
Ω	Input–output matrix	BEA 2017 405-industry input-output table
δ_i	Shipment–inventory ratios	Economic Census 2017 COGS / Inventories
ϕ_i	Weight on accounting costs	Rauch (1999): $\phi_i = 0$ if r/w, $\phi_i = 1$ if n.

- Inventories for mining (21), manufacturing (31–33), wholesale (42) & retail (44–45).
    - Low estimate of historical dependence if firms lock in prices in other ways (e.g., hedging).
- Problem: BEA treats transportation, wholesale, and retail trade as “margin industries.”
    - Construct new “trade-adjusted input–output network” with 843 industries.
    - Reverse engineer flows of goods through trade industries using BEA Margins table.

Inventory–shipment ratios from Economic Census



Industries with largest inventory–shipment ratios:

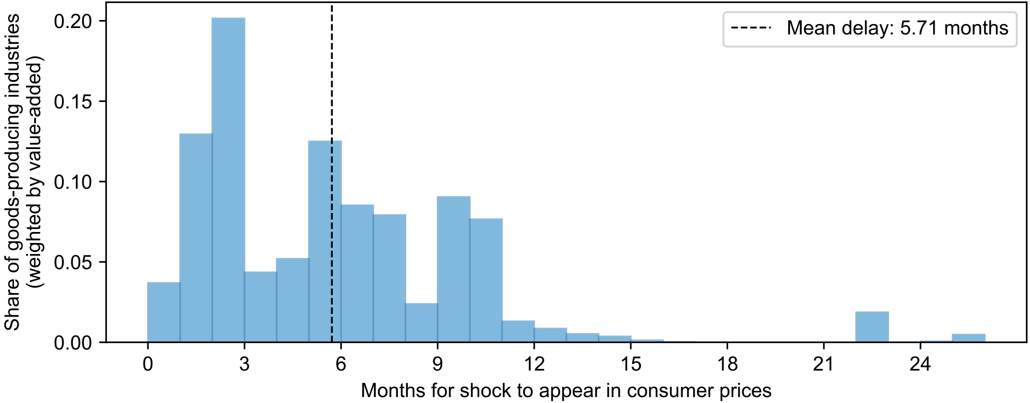
- Wineries.
- Biological product mfg.
- Aircraft mfg.
- Distilleries.
- Aerospace products and parts mfg.

How much rigidity does pricing off accounting costs generate?

1. Delayed effects of cost shocks on CPI.
2. [In paper] Effect of identified oil shocks (Känzig 2021) vs. data.
3. Effect of 2025 tariffs vs. Calvo vs. data.

Long lags between industry cost shocks and CPI effects

- Shock to costs of goods-producing industry takes 5.7 months on average to show up in consumer prices under FIFO (4.9 months under AAC).

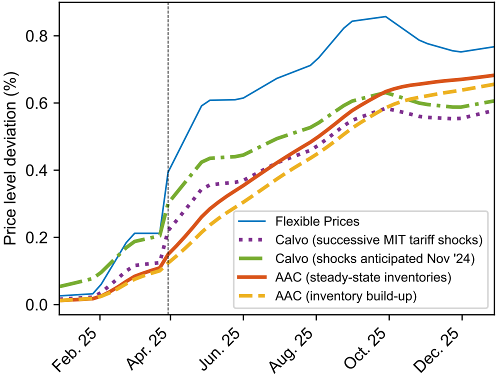


Effect of 2025 tariffs on consumer price index

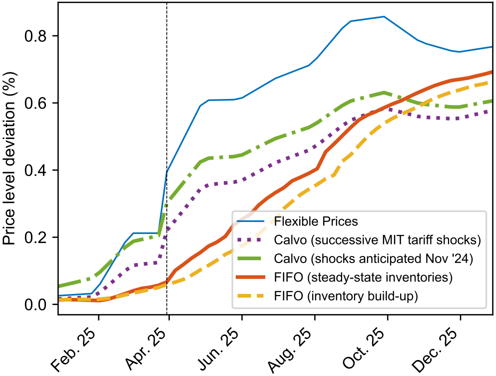
- Monthly effective tariff rates by commodity from Jan ’25 to Jan ’26.
- Two assumptions about inventories:
  1. Inventories remain at steady-state levels.
  2. Stockpiling using import quantities observed in data.
- Also compare to benchmark Calvo model, calibrated á la Rubbo (2023).
  - Standard treatment: Trade industry rigidity only applies to trade margins.
  - Industry-specific price rigidities from Pasten et al. (2020), $\beta = (0.99)^{1/12}$.
  - Expectations: All future tariff changes are unanticipated (baseline).

Effect of 2025 tariffs on consumer price index

- Accounting cost pricing creates delays relative to flex-price benchmark (blue).
- Calvo (á la Rubbo 2023) adjusts more quickly, largely due to lack of trade industries.



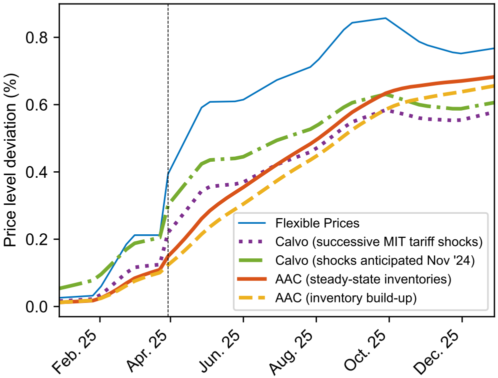
(a) Average acquisition cost (AAC).



(b) First-in-first-out (FIFO).

Effect of 2025 tariffs on consumer price index

- Accounting cost pricing creates delays relative to flex-price benchmark (blue).
- Calvo $+$ anticipation starting in Nov '24 $\rightarrow$ even faster adjustment



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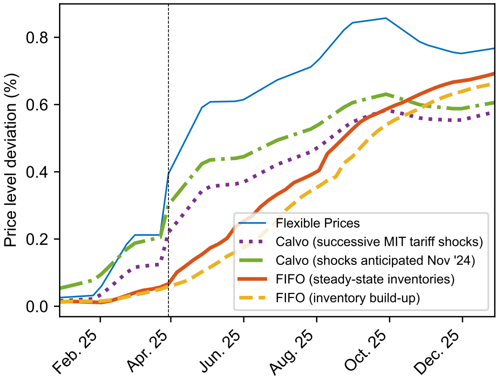
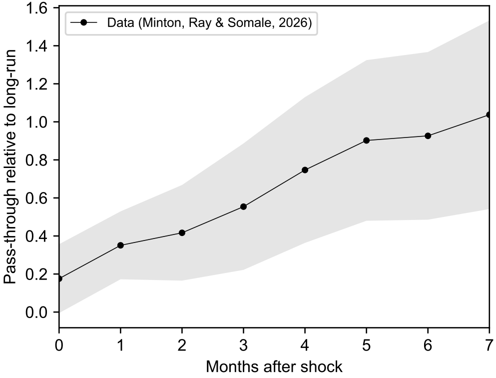


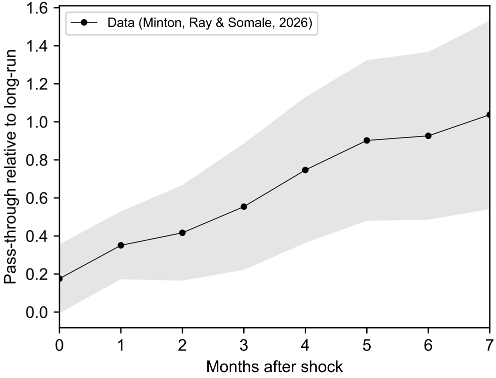
(b) First-in-first-out (FIFO).

Effect of 2025 tariffs on consumer price index: Pass-through delays

- Minton et al. (2026) run distributed lag reg.: $\Delta p_{it} = \sum_{k=0}^{K} \beta_k \Delta p^{\text{LR}}_{it-k} + \alpha_i + \lambda_t + \varepsilon_{it}$.
- Estimate these pass-through regressions in our model-generated data.



(a) Average acquisition cost (AAC).



(b) First-in-first-out (FIFO).

Effect of 2025 tariffs on consumer price index: Pass-through delays

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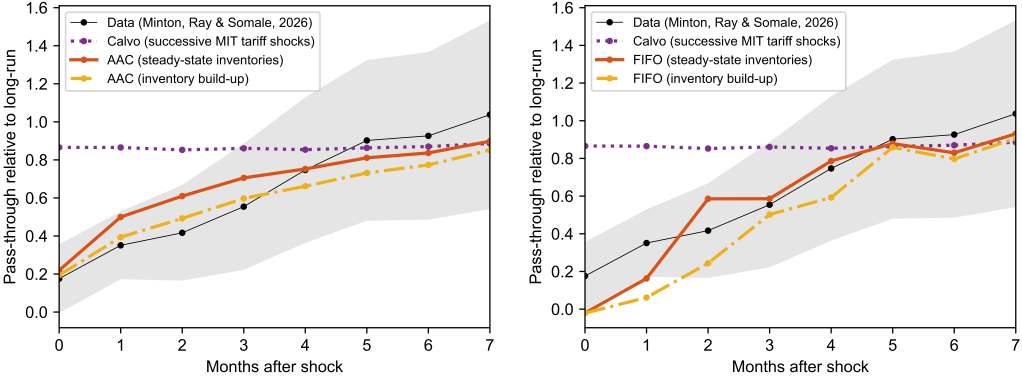

(a) Average acquisition cost (AAC).

(b) First-in-first-out (FIFO).

Effect of 2025 tariffs on consumer price index: Model vs. data

- Accounting cost models appear to capture price responses in data.
- Outperform off-the-shelf Calvo model (largely due to trade industries).

	PCE category monthly inflation: Data				
	(1)	(2)	(3)	(4)	(5)
Inflation in AAC Model	0.782**			1.338**	
	(0.337)			(0.415)	
Inflation in FIFO Model		0.649**			0.666**
		(0.194)			(0.197)
Inflation in Calvo Model			−0.085	−0.574**	−0.155
			(0.124)	(0.204)	(0.110)
Date FEs	Yes	Yes	Yes	Yes	Yes
PCE Category FEs	Yes	Yes	Yes	Yes	Yes
N	6572	6572	6572	6572	6572
R^2	0.04	0.04	0.04	0.04	0.04

Conclusion

- Cost measures used by firms are inherently backward-looking.
- Cost measures respond sluggishly to shocks to replacement costs.
- Pricing off accounting costs thus delays response of prices to replacement costs.
  - Macro price rigidity without sticky prices per se.
  - Delays depend on time it takes firms to work through inputs/inventories.
- Model of accounting cost pricing can also rationalize other pricing puzzles (pass-through of fixed costs, insensitivity to demand shocks).

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The Standard Theory: Firm Problem

- Given inventory I_{t-1} and current replacement cost c_t, firm value is:

$$V(I_{t-1}, c_t) = \max_{p_t, Y_t} \{p_t D_t - c_t Y_t + \beta \mathbb{E}[V_{t+1}(I_t, c_{t+1})]\},$$

where

$$I_t = I_{t-1} - D_t + Y_t. \qquad \text{(Inventory law of motion)}$$

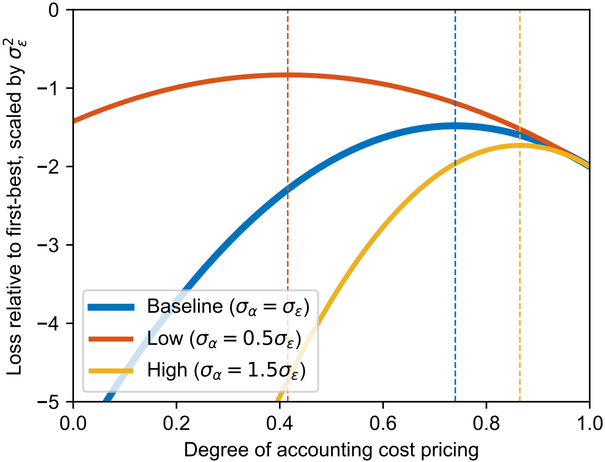
$$mr_t(p_t) = D_t + p_t(dD_t/dp_t). \qquad \text{(Marginal revenue)}$$

- FOCs: Set marginal revenue = marginal cost = current replacement cost:

$$mr_t(p_t) = \beta \mathbb{E}\left[\frac{\partial V_{t+1}}{\partial I_t}\right] = c_t.$$

- Some (unsatisfactory) directions: $Y_t \geq 0$, $D_t \leq I_{t-1}$, perishable inventory, convex inventory carrying costs, borrowing constraint.

Losses relative to first best



- With optimal contract and $\tau = 0$,

  $$\mathscr{L} = \mathbb{E}[V^{\mathrm{FB}} - V] \approx (\eta - 1)(1 - \phi)\sigma_c^2.$$

- If $\sigma_c^2 = 0$, $\mathscr{L} = 0$.

- If $\sigma_\alpha^2 = 0$, $\phi = 1$ and $\mathscr{L} = 0$.

- $\mathscr{L}$ nonmonotonic in η.
  ($\uparrow$ losses but $\phi \to 1$.)

Alternative explanations

- **Earnings smoothing.** Graham et al. (2005), Terry (2023).
  - Earnings depend on quantities $\Rightarrow$ depends on what other firms do.
  - If quantities don't change in equilibrium (e.g., all firms set same price and industry demand inelastic), then earnings smoothing $=$ setting target accounting markup.
- **Fairness.** Kahneman et al. (1986), Rotemberg (2005, 2011), Eyster et al. (2021).
  - Similar structure: outsider tries to evaluate whether price is justified.
  - Fairness binds more on price increases than decreases. But estimates of ϕ symmetric.

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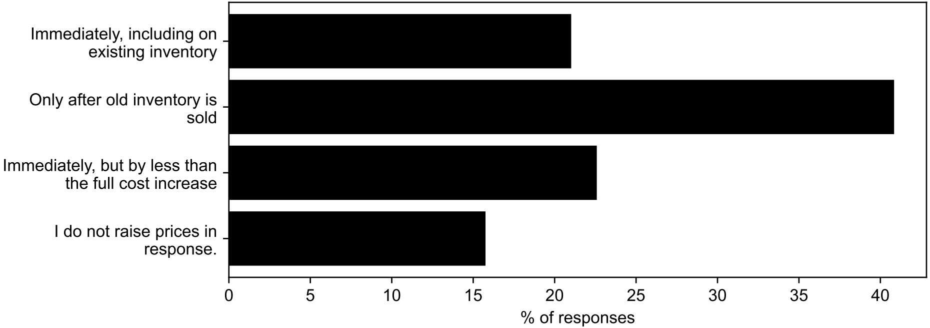
Survey evidence

1. Many firms report waiting to raise prices until inventory at old cost is exhausted.
2. Pass-through of changes in overhead costs.
3. Limited pass-through of expected future costs.

Survey evidence

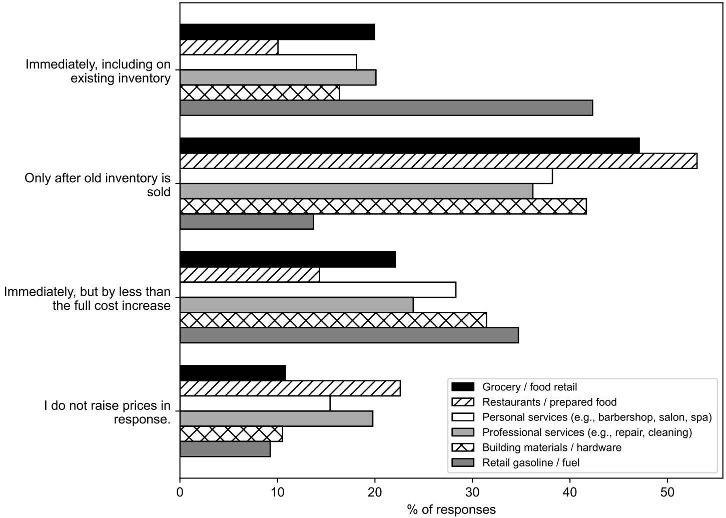
- “Suppose your input cost rises today, but you still have inventory bought at lower cost. When do you raise your sales price?”



Note: Survey of 400 firm price-setters by Schoenle (2026).

- Not about uncertainty about future costs!

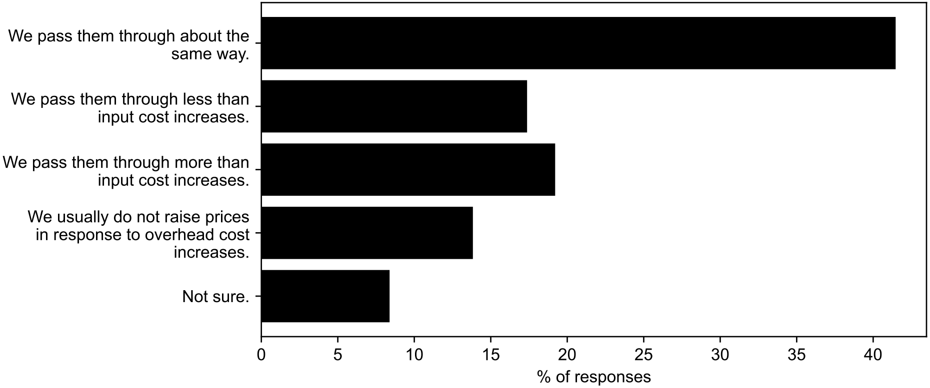
Exceptions



- Exception: gas stations.
- "The exceptions that occur to me arise when processed raw materials are sold on auction markets, as in the case of flour or soybean meal." (Okun 1981).
- (Anecdotal) exception: sellers of gold/silver jewelry.

Firms' reported pass-through of fixed costs

- "Suppose your overhead costs go up, for example on rent, insurance, salaried staff or equipment leases. How do increases in such overhead costs affect your pricing—compared to increases in per-unit input costs on materials, wholesale goods, shipping per item, or hourly labor per job?"



Prices don't incorporate expected cost changes, but not b/c firms uninformed

- Fed Atlanta BIE survey: Firms' price changes incorporate realized, past cost changes but not expectations of future cost changes.
- Not because firms inattentive: Expectations predict future cost changes.

	Realized Price Change$_t$		*Realized Cost Change$_{t+12}$*	
	(1)	(2)	(3)	(4)
$\mathbb{E}_t$[Cost Change$_{t+12}$]	-0.082	-0.169	0.354**	0.294**
	(0.204)	(0.182)	(0.090)	(0.083)
Realized Cost Change$_t$	1.272**	0.979**	-0.050	-0.040
	(0.305)	(0.233)	(0.055)	(0.048)
Firm FEs	Yes	Yes	Yes	Yes
Date FEs	Yes		Yes	
Date-Sector FEs		Yes		Yes
N	3493	3480	1415	1410
R^2	0.48	0.56	0.65	0.73

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Can we estimate the extent of accounting cost pricing?

- Suppose price $=$ markup $\times$ cost.

- Cost measure puts weight ϕ on replacement cost c_t, $1-\phi$ on accounting cost h_t.

$$p_t = \mu\left(\phi c_t + (1-\phi)h_t\right).$$

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$$p_t = \mu\left(\phi c_t + (1-\phi)h_t\right).$$

- Accounting markup (sales / cost of goods sold) is:

$$\frac{\text{Sales}_t}{\text{COGS}_t} = \frac{\mu(\phi c_t + (1-\phi)h_t)D_t}{h_t D_t} = \mu\left(\phi\frac{c_t}{h_t} + (1-\phi)\right).$$

- $\phi > 0 \Rightarrow$ prices high relative to accounting cost when $c_t > h_t$.

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- $\phi > 0 \Rightarrow$ prices high relative to accounting cost when $c_t > h_t$.
- How can we tell when $c_t > h_t$?

Can we estimate the extent of accounting cost pricing?

- Key insight: Purchases capitalized into inventory.

$$\text{BookValueInventory}_t = \text{BookValueInventory}_{t-1} - \underbrace{\text{COGS}_t}_{h_t D_t} + \underbrace{\text{Purchases}_t}_{c_t Y_t}.$$

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- Thus,

$$\frac{\text{Purchases}_t}{\text{COGS}_t} = \frac{\Delta\text{BookValueInventory}_t + \text{COGS}_t}{\text{COGS}_t} = \frac{c_t Y_t}{h_t D_t}.$$

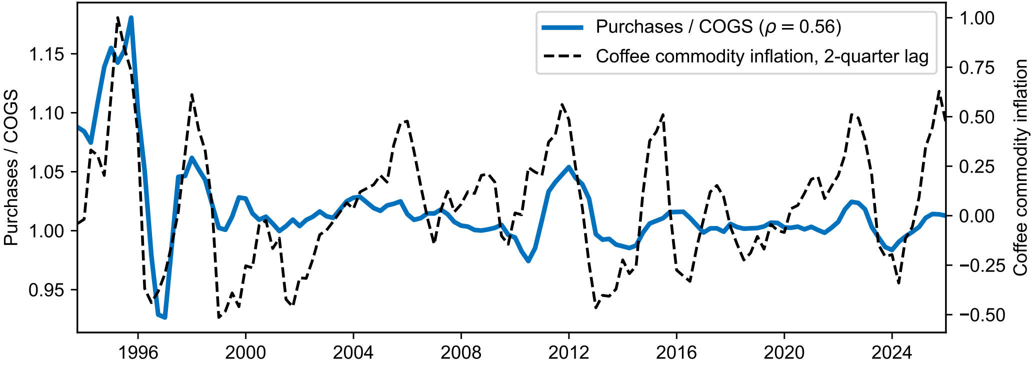
- Log-linearizing:

$$\widehat{\left(\frac{\text{Sales}_t}{\text{COGS}_t}\right)} = \phi \widehat{\left(\frac{\text{Purchases}_t}{\text{COGS}_t}\right)} - \phi\left(\hat{Y} - \hat{D}\right).$$

- Absent changes in quantities and markups, ϕ identified by sensitivity of Sales/COGS to Purchases/COGS.

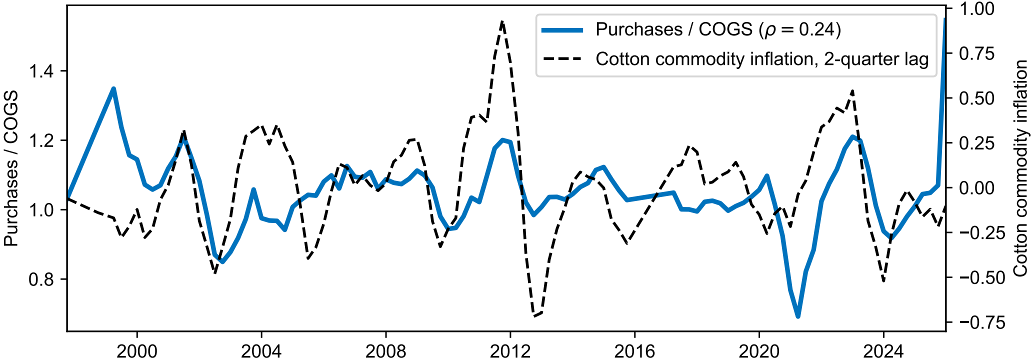
Does Purchases/COGS capture variation in c_t/h_t? Example: Starbucks

- Use 10Qs for sales, COGS, purchases $=$ ΔBookValueInventory $+$ COGS.
- Compare to ICO Arabica coffee bean commodity price.



Does Purchases/COGS capture variation in c_t/h_t? Example: Gildan

- Use 10Qs for sales, COGS, purchases $=\Delta$BookValueInventory$+$COGS.
- Compare to Cotlook benchmark cotton spot price.



Estimating ϕ across all firms

- Quarterly financial reports for public firms:
  - Exclude firms in information (NAICS 51), finance/insurance/real estate (NAICS 52–53).
  - Observations with strictly positive sales, COGS, and inventory stock.
  - Firms using AAC or FIFO. (Under LIFO, $h_t \approx c_t$.)
  - Change in total assets over past year $< 30\%$ (avoid mergers / spin-offs).
- Specification:

$$\log(\text{Sales}/\text{COGS})_{it} = \phi \log(\text{Purchases}/\text{COGS})_{it} + \alpha_{iq} + \varepsilon_{it}.$$

where α_{iq} are firm $\times$ quarter-of-year FEs.

Estimating ϕ across all firms

- Specification:

$$\log(\text{Sales}/\text{COGS})_{it} = \phi \log(\text{Purchases}/\text{COGS})_{it} + \alpha_{iq} + \varepsilon_{it}.$$

- Change in relative quantities of purchases / shipments $\hat{Y}_t - \hat{D}_t$ is omitted variable:

$$\hat{\phi} \to \phi \left[1 \underbrace{- \frac{Var\left(\hat{Y}_t - \hat{D}_t\right)}{Var\left(\widehat{\frac{\text{Purchases}}{\text{COGS}}}\right)}}_{\substack{\text{Attenuation bias} \\ <0}} + \underbrace{\frac{-Cov\left(\hat{c}_t - \hat{h}_t, \hat{Y}_t - \hat{D}_t\right)}{Var\left(\widehat{\frac{\text{Purchases}}{\text{COGS}}}\right)}}_{\substack{\text{Forward-buying bias} \\ \text{likely } >0}} \right].$$

  - Seasonal variation in $\hat{Y}_t - \hat{D}_t$: firm $\times$ quarter-of-year FEs.
  - Attenuation bias: IV using NAICS-4 industry $\times$ Känzig (2021) oil shocks.
  - IV estimates are upper bound if firms engage in forward buying ($\hat{Y}_t \downarrow$ when $\hat{c}_t \uparrow$).

Estimating ϕ: All firms

- OLS: $\phi \in [0.12, 0.15]$.
- NAICS-4 $\times$ Känzig (2021) oil shock IV: $\phi \in [0.35, 0.4]$.
- Extent of accounting cost pricing $(1 - \phi) \approx$ 60–80%.

	Log(Sales / COGS)			
	Levels		First Differences	
	OLS	IV	OLS	IV
	(1)	(2)	(3)	(4)
Log(Purchases / COGS)	0.147**	0.380**	0.126**	0.351**
	(0.015)	(0.104)	(0.014)	(0.079)
Firm $\times$ Quarter-of-Year FEs	Yes	Yes	Yes	Yes
N	382 211	369 469	345 468	333 935
R^2	0.86	0.86	0.09	0.04

More weight on recent costs when $\uparrow$ turnover

- h_t should move more quickly toward c_t when inventory turns over faster.
- Weight on replacement costs $\uparrow$ as firm inventory turnover $\delta = \text{COGS}/\text{Inventories} \uparrow$.

	Log(Sales$_{t\to t+4}$/COGS$_{t\to t+4}$)			
	Levels		First Differences	
	(1)	(2)	(3)	(4)
Log(Purchases$_{t\to t+4}$/COGS$_{t\to t+4}$)	0.087		0.087**	
	(0.078)		(0.025)	
Log(Purchases$_{t\to t+4}$/COGS$_{t\to t+4}$) $\times$ log(δ)	0.162**	0.154**	0.063**	0.065**
	(0.073)	(0.078)	(0.028)	(0.028)
Firm $\times$ Quarter-of-Year FEs	Yes	Yes	Yes	Yes
NAICS-2 FEs $\times$ Log(Purchases$_{t\to t+4}$/COGS$_{t\to t+4}$)		Yes		Yes
N	347 829	347 829	320 894	320 894
R^2	0.89	0.89	0.11	0.11

Estimating ϕ: Comparative statics

1. **Homogeneous outputs or observable costs.**
   - Rauch (1999): Industries with reference prices (*r*) or organized exchanges (*w*) vs. administered prices (*n*).
2. **Competitors pricing off replacement costs.**
   - Share of firms in industry using LIFO in industry from Economic Census.

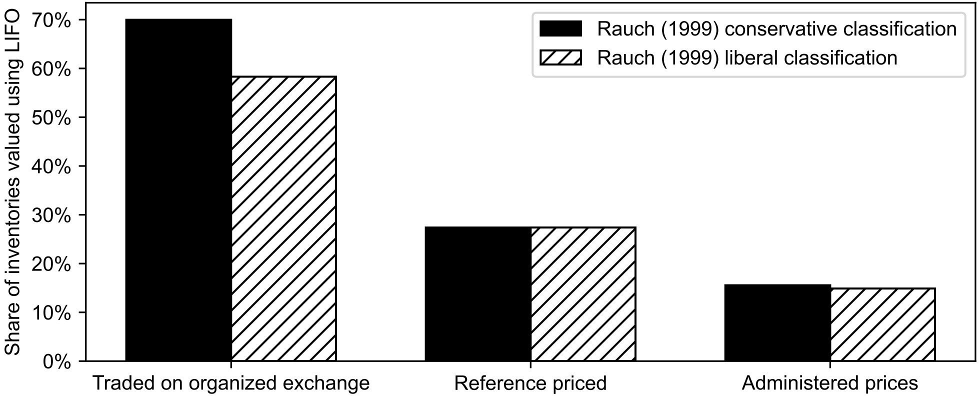
Estimating ϕ: Comparative statics

- $\uparrow \phi$ with reference prices / organized exchanges, or where competitors use LIFO.

	Log(Sales / COGS)					
	Levels		First Differences			
	OLS	OLS	OLS	OLS	IV	IV
	(1)	(2)	(3)	(4)	(5)	(6)
Log(Purchases / COGS)	0.122**	0.125**	0.101**	0.103**	−0.268	−0.136
	(0.016)	(0.019)	(0.017)	(0.016)	(0.376)	(0.265)
× Reference prices (r)	0.079**		0.074*		1.006*	
	(0.029)		(0.042)		(0.599)	
× Organized exchange (w)	0.254**		0.168**		0.720**	
	(0.062)		(0.051)		(0.329)	
× Industry LIFO share		0.185**		0.144**		0.744**
		(0.090)		(0.068)		(0.353)
Firm × Quarter-of-Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
N	251 379	276 804	227 964	251 033	225 546	242 882
R^2	0.88	0.88	0.09	0.09	0.08	0.01

AAC/FIFO common when accounting cost pricing likely

- LIFO concentrated in industries with homogeneous outputs (organized exchanges) or inputs (reference-priced).



Summary of estimates

- Average extent of accounting cost pricing $(1-\phi) \approx$ 60–80%.
  - Accounting cost pricing $\geq$ 85% for firms in industries with differentiated products (*n*).
  - Replacement cost pricing in industries with reference prices (*r*) or exchanges (*w*).
- Dependence on historical costs not due to inattention to replacement costs.
  - Test uses purchases already made by firm!

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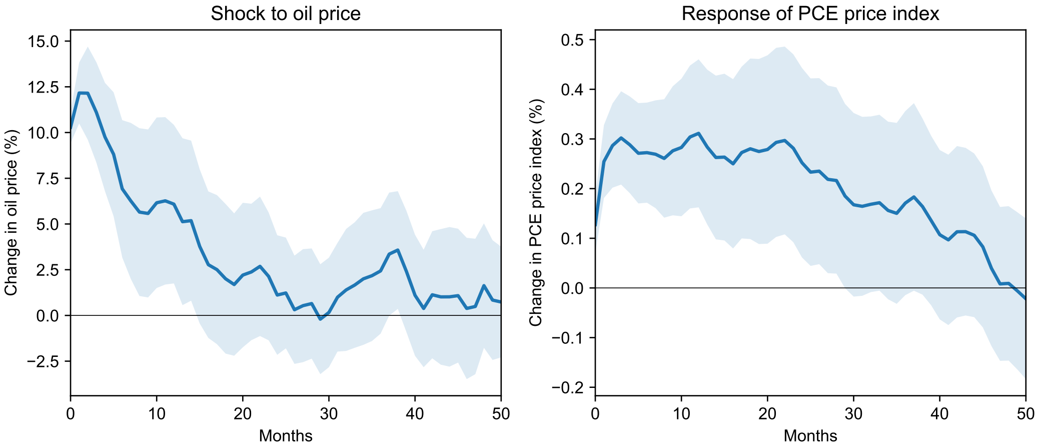
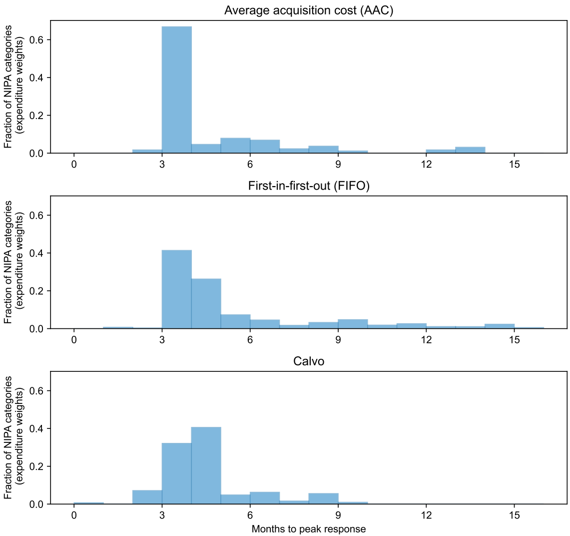
Effect of oil shocks

- Feed in oil news shocks isolated from OPEC announcements by Känzig (2021).



Effect of oil shocks: Models predict delayed impact on consumer prices



- Oil shock peaks in month 1.
- Models with accounting cost pricing predict peak responses of most consumer prices around 3–6 months.
- Similar to Calvo despite flex prices!

Effect of oil shocks: Model vs. data

- Measure impulse response of each PCE subcomponent to oil shocks.
- Accounting cost models in line with data. Similar performance to Calvo.

	Response of PCE category price index to oil shock: Data				
	(1)	(2)	(3)	(4)	(5)
Price Response in AAC Model	1.066**			0.954**	
	(0.171)			(0.382)	
Price Response in FIFO Model		1.042**			0.465**
		(0.161)			(0.131)
Price Response in Calvo Model			1.060**	0.113	0.616**
			(0.173)	(0.408)	(0.226)
N	10812	10812	10812	10812	10812
R^2	0.09	0.09	0.09	0.09	0.09

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